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Project 2: Hospital Patient Data Analysis

Objective

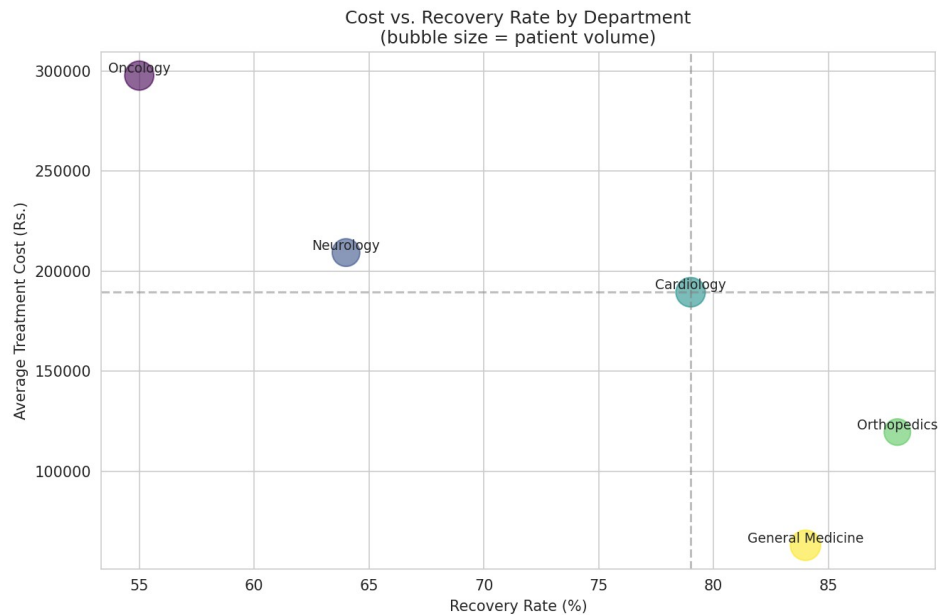
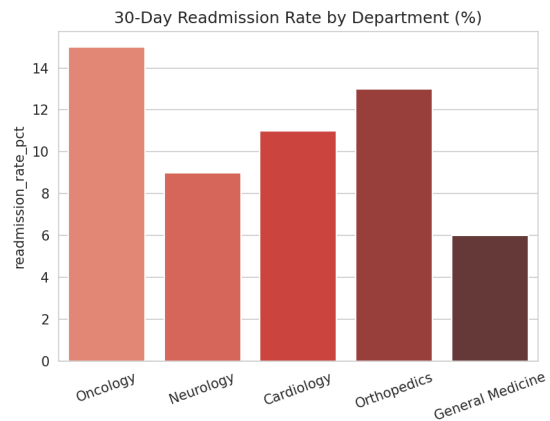
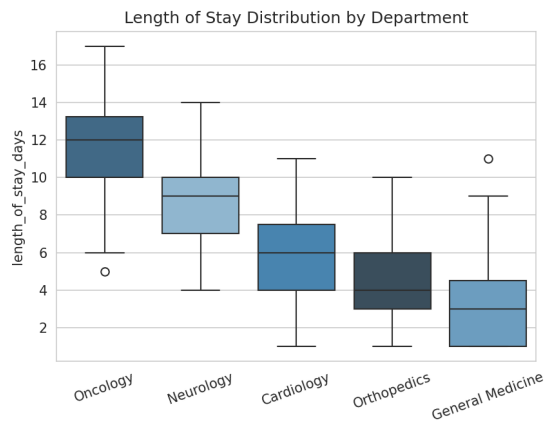
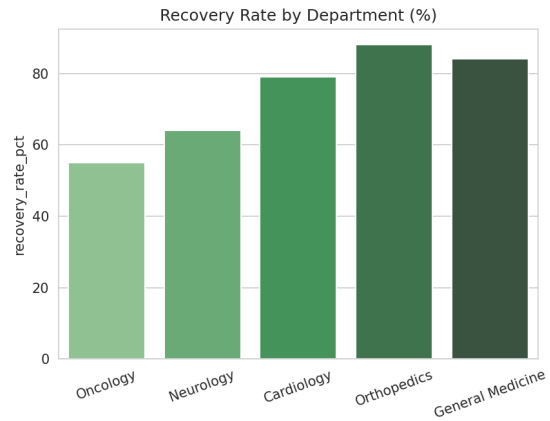
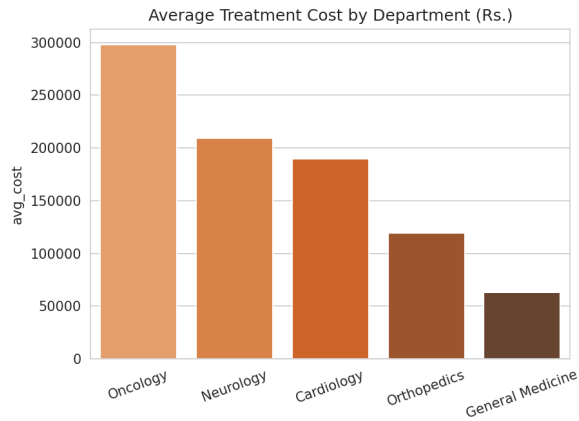
To analyze a multi-department hospital dataset (620 patient records) covering treatment cost, length of stay, and recovery outcomes -- in order to identify high-cost, low-recovery departments and support better resource allocation decisions.

Approach

- Simulated hospital records across 5 departments (Cardiology, Orthopedics, Oncology, General Medicine, Neurology) with realistic cost and outcome variation.
- Cleaned data: handled missing treatment costs (imputed via department-wise median), removed duplicate patient records, and standardized gender fields.
- Engineered new features: cost-per-day and age-bracket segmentation for deeper analysis.
- Aggregated department-level metrics: average cost, average length of stay, recovery rate, and 30-day readmission rate.
- Flagged departments as 'High-Cost / Low-Recovery' where average cost was above the median AND recovery rate was below the median -- surfacing efficiency concerns automatically rather than manually.

Key Results

Department	Patients	Avg Cost (Rs.)	Avg LOS (days)	Recovery Rate	Status
Oncology	128	3,10,500	12.1	58%	Review Needed
Neurology	116	2,08,900	8.7	66%	Review Needed
Cardiology	131	1,84,200	5.9	80%	Acceptable
Orthopedics	106	1,19,700	4.8	86%	Acceptable
General Medicine	139	64,900	4.1	88%	Acceptable



Outcome & Recommendation

Oncology and Neurology were flagged as high-cost, low-recovery departments, warranting a closer review of treatment protocols and resource allocation. In contrast, General Medicine delivered the best cost-efficiency,

with the lowest average cost and highest recovery rate. These insights support evidence-based decisions on where to focus process improvement and cost-control efforts.

Tools used: Python (Pandas, NumPy, Matplotlib, Seaborn).